

Paper 1: Medicine Through Time with the Western Front

Key Events Timeline

c1250 - Medieval Medicine

Medicine is dominated by the Catholic Church and the ancient theories of Galen (Four Humours). Astrology and religion are the main explanations for disease.

1348 - The Black Death arrives in England

A devastating outbreak of bubonic plague kills roughly one-third of England's population. People blame God, miasma, and the planets.

1543 - Vesalius publishes 'On the Fabric of the Human Body'

Andreas Vesalius proves Galen made over 300 anatomical mistakes, encouraging other doctors to dissect and observe for themselves.

1628 - William Harvey proves the circulation of blood

Harvey publishes his findings showing that blood is pumped around the body by the heart, disproving Galen's theory that blood was constantly made in the liver.

1665 - The Great Plague

London is hit by another massive plague outbreak. Local governments try to impose quarantines, but 100,000 people still die.

1676 - Thomas Sydenham publishes 'Observationes Medicae'

The 'English Hippocrates' encourages doctors to observe patient symptoms closely rather than relying on ancient books.

1796 - Edward Jenner discovers the smallpox vaccine

Jenner proves that infecting a person with cowpox protects them from the deadly smallpox virus, leading to the world's first vaccine.

1847 - James Simpson discovers chloroform

An effective anesthetic is discovered, finally allowing surgeons to perform longer and more complex operations without the patient feeling pain.

1854 - John Snow's Broad Street Pump mapping

Snow proves that cholera is spread by contaminated water rather than miasma, though the government is slow to listen.

1861 - Louis Pasteur publishes Germ Theory

Pasteur proves that microbes in the air cause decay, fundamentally changing the understanding of what causes disease.

1865 - Joseph Lister pioneers antiseptic surgery

Lister begins using carbolic acid to kill bacteria during surgery, dramatically reducing death rates from infection.

1875 - Second Public Health Act

The government is forced to take responsibility for public health, forcing cities to provide clean water and proper sewers.

1882 - Robert Koch identifies the Tuberculosis bacteria

Koch builds on Pasteur's work by identifying the specific microbes that cause deadly diseases.

1895 - Wilhelm Roentgen discovers X-rays

Roentgen accidentally discovers X-rays, allowing doctors to look inside the human body without cutting it open.

1901 - Karl Landsteiner discovers blood groups

Landsteiner discovers blood groups (A, B, O), solving the mystery of why previous blood transfusions often killed the patient.

1909 - Paul Ehrlich discovers Salvarsan 606

The first 'magic bullet' is discovered—a chemical compound that targets and kills syphilis bacteria without harming the patient.

1914 - Outbreak of WWI: RAMC and FANY

World War I begins. The Royal Army Medical Corps and First Aid Nursing Yeomanry deploy to the Western Front to treat massive casualties.

1915 - The Thomas Splint

Introduced by Hugh Owen Thomas and championed by his nephew Robert Jones, this splint dramatically increased survival rates for soldiers with shattered femurs.

1915 - First use of Poison Gas

Chlorine gas is used for the first time at the Second Battle of Ypres, leading to the rapid development of gas masks.

1917 - First Blood Bank at Cambrai

Oswald Hope Robertson sets up the first blood bank using sodium citrate to stop blood clotting, saving countless lives at the Battle of Cambrai.

1917 - Harold Gillies and Plastic Surgery

Gillies sets up a specialized hospital at Queen's Hospital, Sidcup, to pioneer plastic surgery techniques for soldiers with severe facial wounds.

1928 - Alexander Fleming discovers Penicillin

Fleming accidentally discovers that a specific mold kills staphylococcus bacteria.

1932 - Gerhard Domagk discovers Prontosil

The second 'magic bullet' is discovered, a red dye that cures blood poisoning (bacteriostatic antibiotic).

1940 - Florey and Chain purify Penicillin

The Oxford team successfully purifies penicillin, leading to mass production during WWII.

1948 - The NHS is established

Aneurin Bevan launches the National Health Service, providing free medical care at the point of need for everyone in Britain.

1953 - The structure of DNA is discovered

Watson, Crick, Franklin, and Wilkins discover the double-helix structure of DNA, opening the door to genetics.

2003 - Human Genome Project Completed

Scientists successfully map every single gene in the human body, allowing doctors to better predict and treat genetic diseases.

2007 - UK Smoking Ban

The government makes it illegal to smoke in all enclosed public spaces and workplaces to combat lung cancer caused by second-hand smoke.

2016 - Plain Packaging Laws

All cigarettes in the UK must be sold in standardized, unappealing green packaging with graphic health warnings to deter young people from smoking.

2020 - COVID-19 Pandemic & mRNA Vaccines

The world faces a deadly pandemic. Thanks to massive government funding and modern technology, highly effective mRNA vaccines are developed and rolled out in under a year.

2024 - Smoking & Vaping Restrictions

The UK government pushes legislation to create a 'smoke-free generation' by banning the sale of tobacco to anyone born after 2009, alongside strict new regulations on disposable vapes.